

Introduction

Aerospace Group Design Project CADEM0016

Unit Director: Fintan Healy (QB1.31) fintan.healy@bristol.ac.uk

Notes for 2026

- Add talk of peer assessment into this presentation
- Think about TLARs a bit more, story wasn't coherent

Unit Structure

1

2

Specification Overview

Getting Started

3

Essential Details

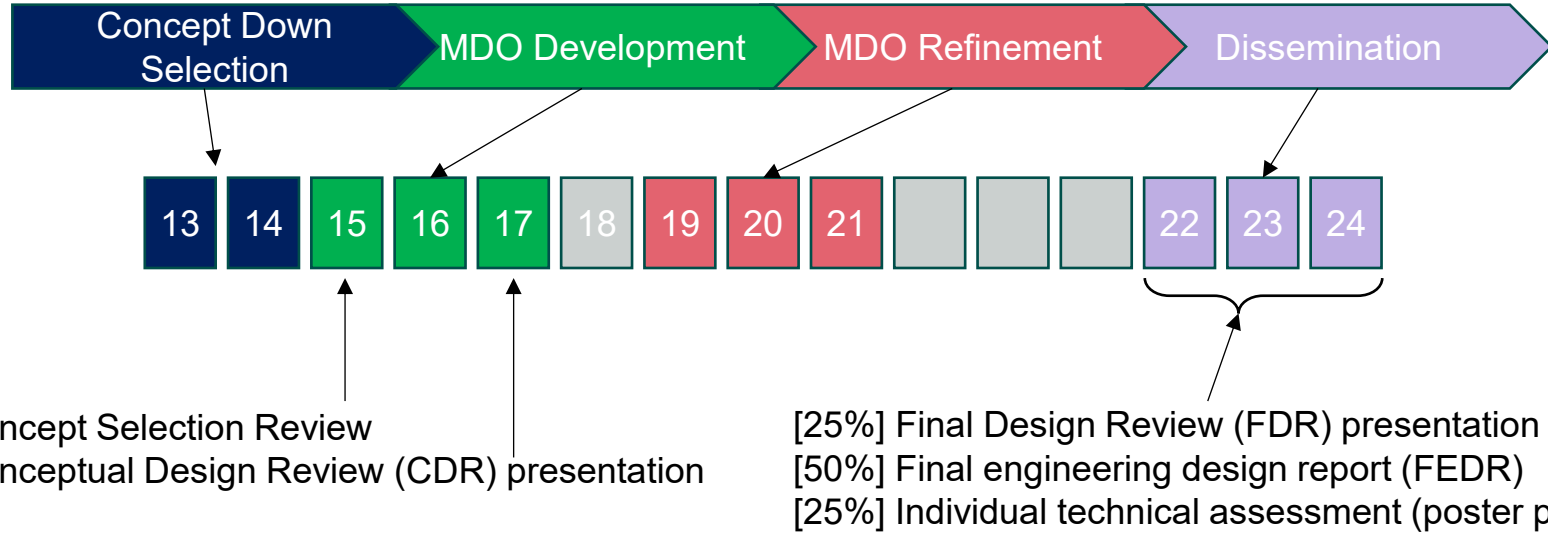
- CADEM0016
- Group Design Project – keystone project
 - Integrator of Learning
- 20 Credits
- Unit Director: Fintan Healy*
- Each student will:
 - Work as part of a group to complete the conceptual design
 - Work on an individual specialism with the design process



ILOs

- **interpret design requirements** and propose appropriate conceptual solutions for a multidisciplinary aerospace engineering design brief
- **apply analysis methods** and tools to refine a selected design iteratively
- apply **teamwork** and project management skills for collaborative efforts to satisfy a design specification, and effectively integrate contributions from multiple technical disciplines
- effectively **present design processes and decisions**, and judiciously defend them under scrutiny for multiple disciplines

Overall Process & Key Dates

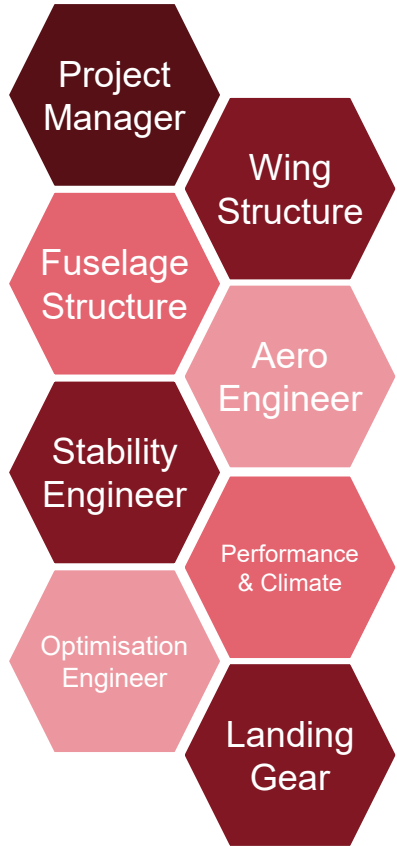


Schedule

- 20 credits
 - 200 hours of work
 - ~2 days a week (16 hrs)
- Two lecture slots & one group session each week
 - 5 hours a week
- ~11 hours a week self/group work
 - You can meet as a group outside of these times!

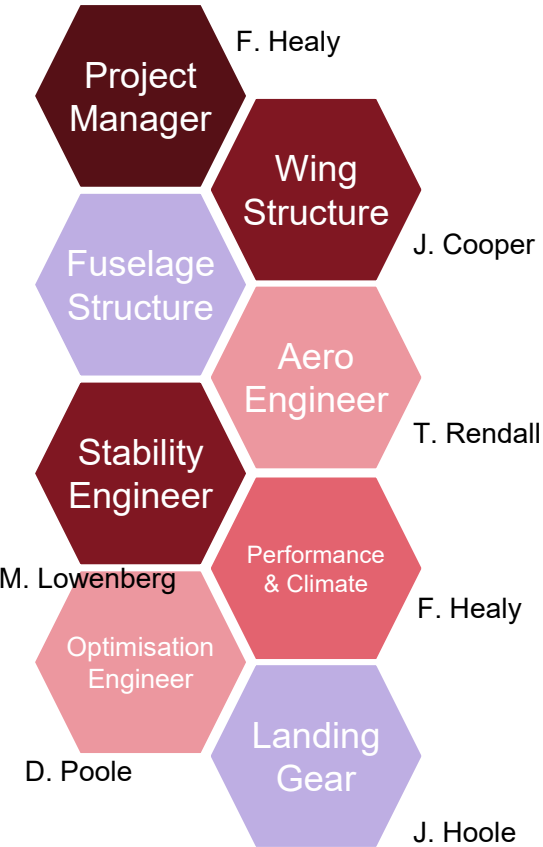
Weeks	W.C.	Monday 10am-11am	Wed 11am-12pm	Friday 2pm-3pm	3pm-5pm
13	19/01	Dr Healy - Welcome Talk	Dr Healy - Conceptual Design 101	Dr Healy - Conceptual Design 102	Group Work
14	26/01	Dr Rendall - Aerodynamics	Prof. Cooper - Structures	Dr Healy - Introduction to MDO	Project Management Skills + Group Work
15	02/02	Prof. Lowenberg - Stability & Control	Dr Healy - Aircraft Economics	Dr Poole - Optimisation	Group Work
16	09/02	Dr Hoole - Landing Gear	Dr Healy - Climate Impact 1	Dr Healy - CDR Overview	Group Work
17	16/02	-	-	CDR	CDR
18	23/02	-	-	Group Work	Group Work
19	02/03	-	-	Group Work	Group Work
20	09/03	-	-	Group Work	Group Work
21	16/03	-	-	Dr Healy - FDR, FEDR & Poster Session Overview	Group Work
22	13/04	-	-	FDR	FDR
23	20/04	-	-	-	-
24	27/04	-	-	Poster Presentation	Poster Presentation

Team Roles



- Teams of 7 or 8 students
 - See Blackboard / “Teams” for your group allocations.
- For sprint 1 (Concept Down Selection), you work on a common task.
- But you then specialise into 1 of 8 roles.

Team Roles

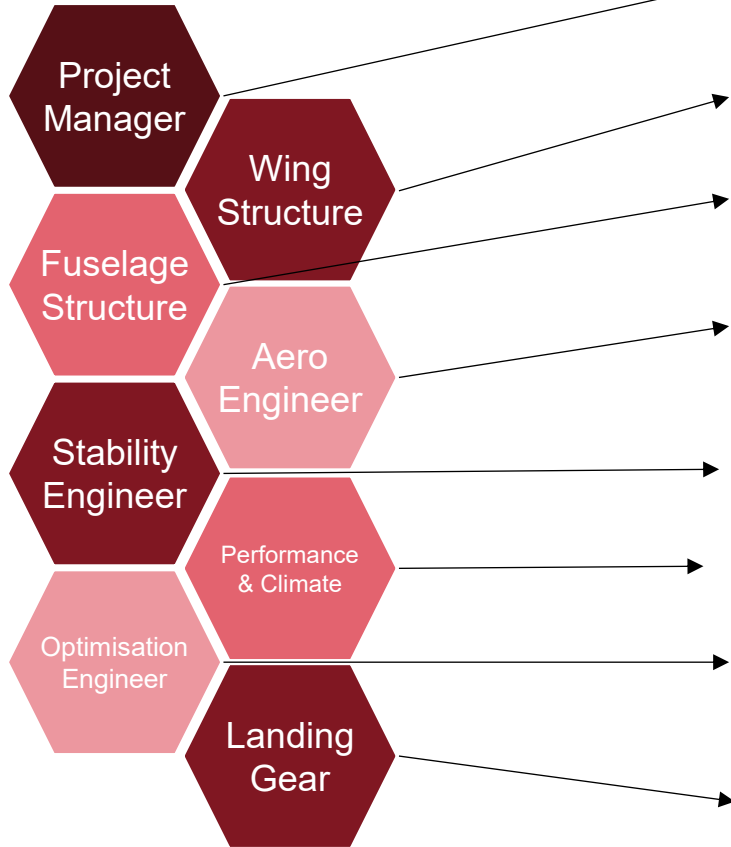


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Note

Teams of 7 can pick between the Fuselage and Landing Gear Engineer

Team Roles



Project Manager & Economics Engineer:

- Overall coordination, document management
- Economic Analysis discipline (Direct operating cost model)

Wing Structures Engineer:

- Wingbox design, loads analysis

Fuselage Structures Engineer:

- Fuselage and Empennage structural design & loads analysis

Aerodynamics Engineer:

- Wing planform & aerofoil selection
- Drag Polars
- High-lift devices

Empennage & Stability Engineer:

- Tail sizing, static/manoeuvre margins
- CG range, trim drag

Performance, Propulsion & Climate Engineer:

- Engine selection, mission analysis, climate impact analysis

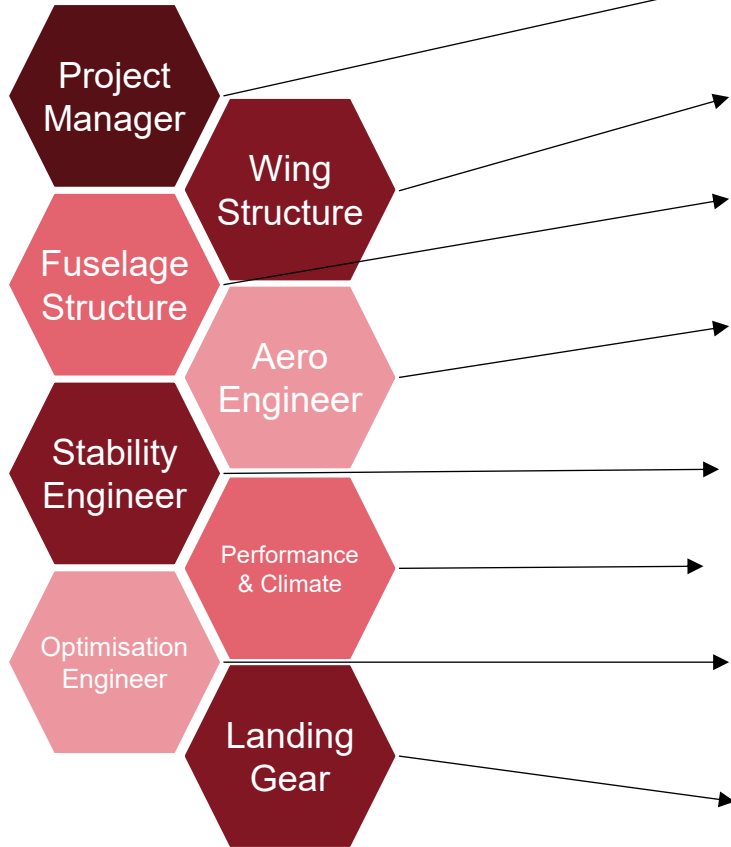
Optimisation and Integration Engineer:

- System-level optimisation, trade studies, design freeze coordination & sensitivity analysis

Landing Gear Engineer:

- Gear layout, footprint, shock absorber design, stowage, and integration

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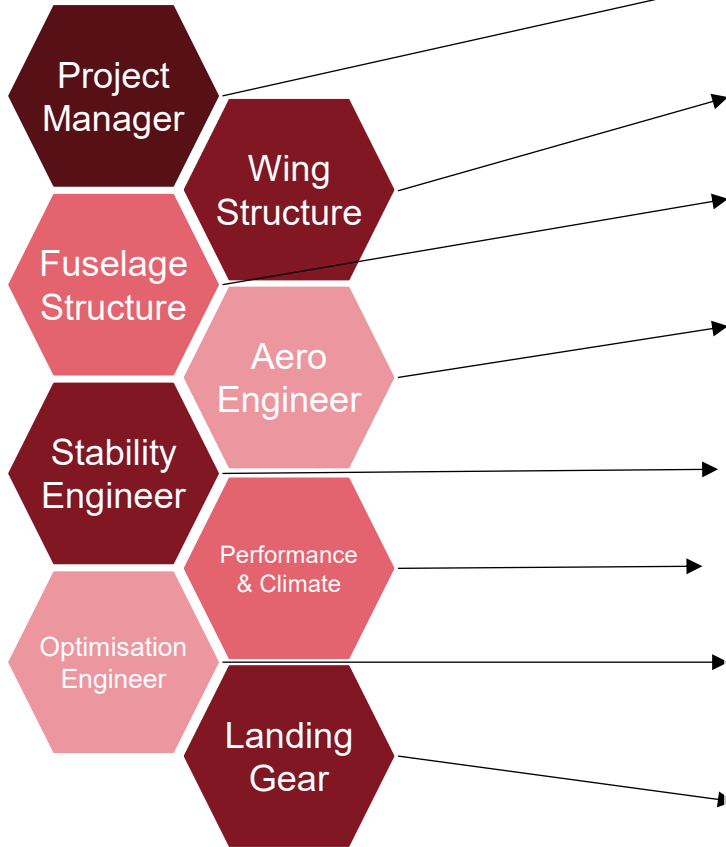
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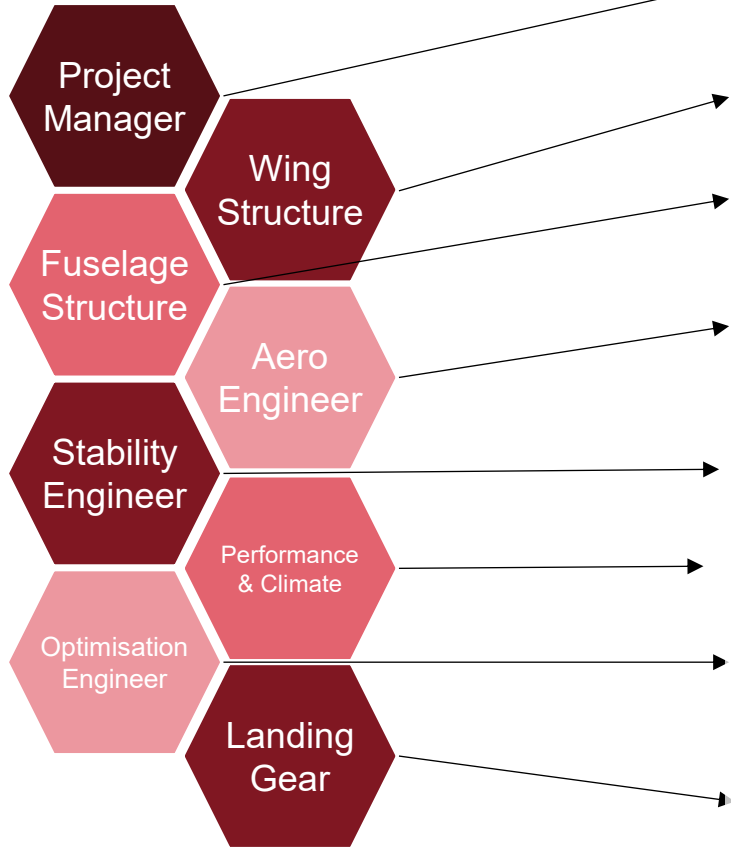
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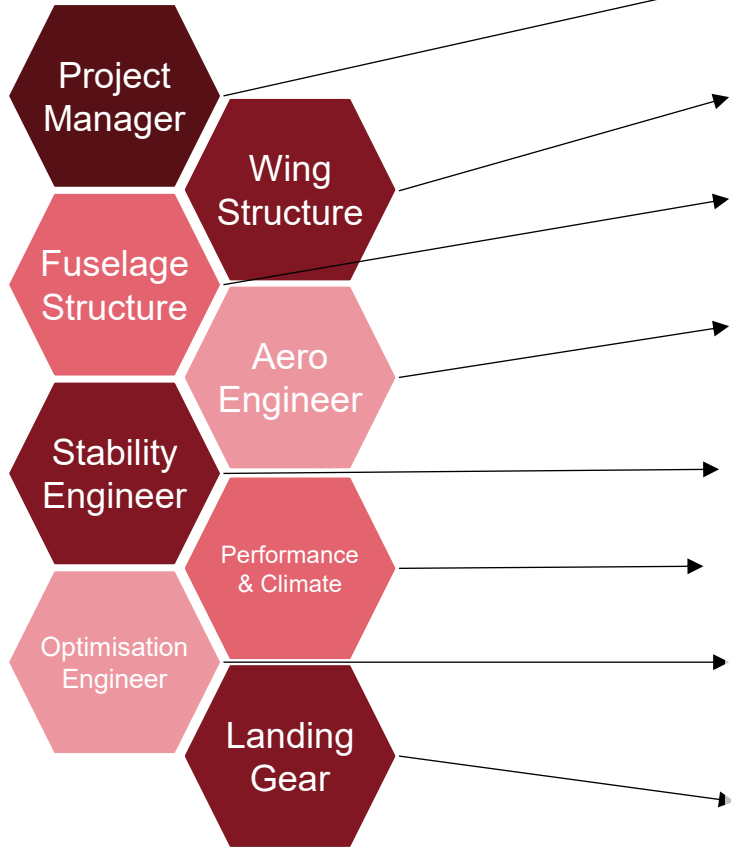
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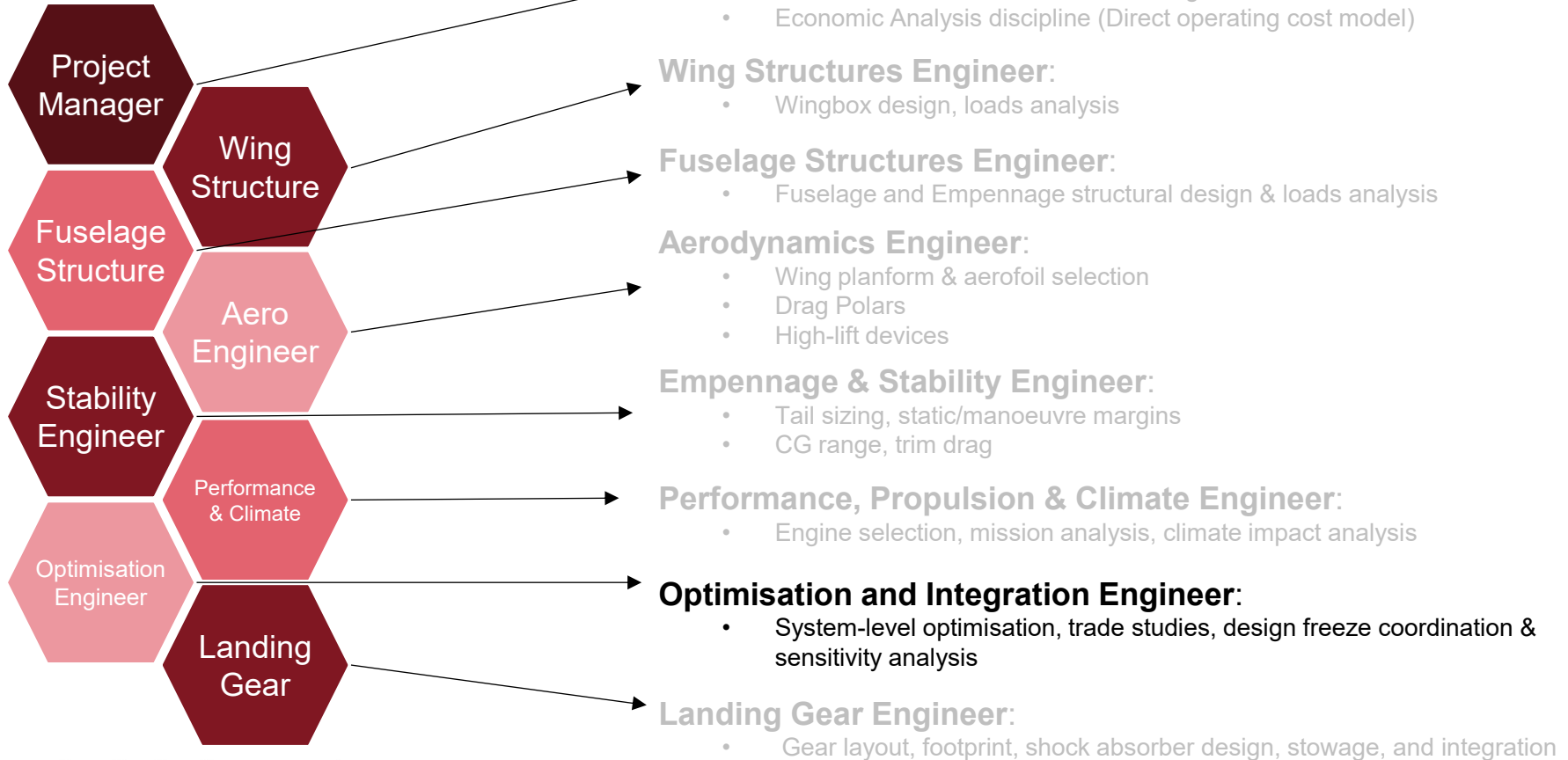
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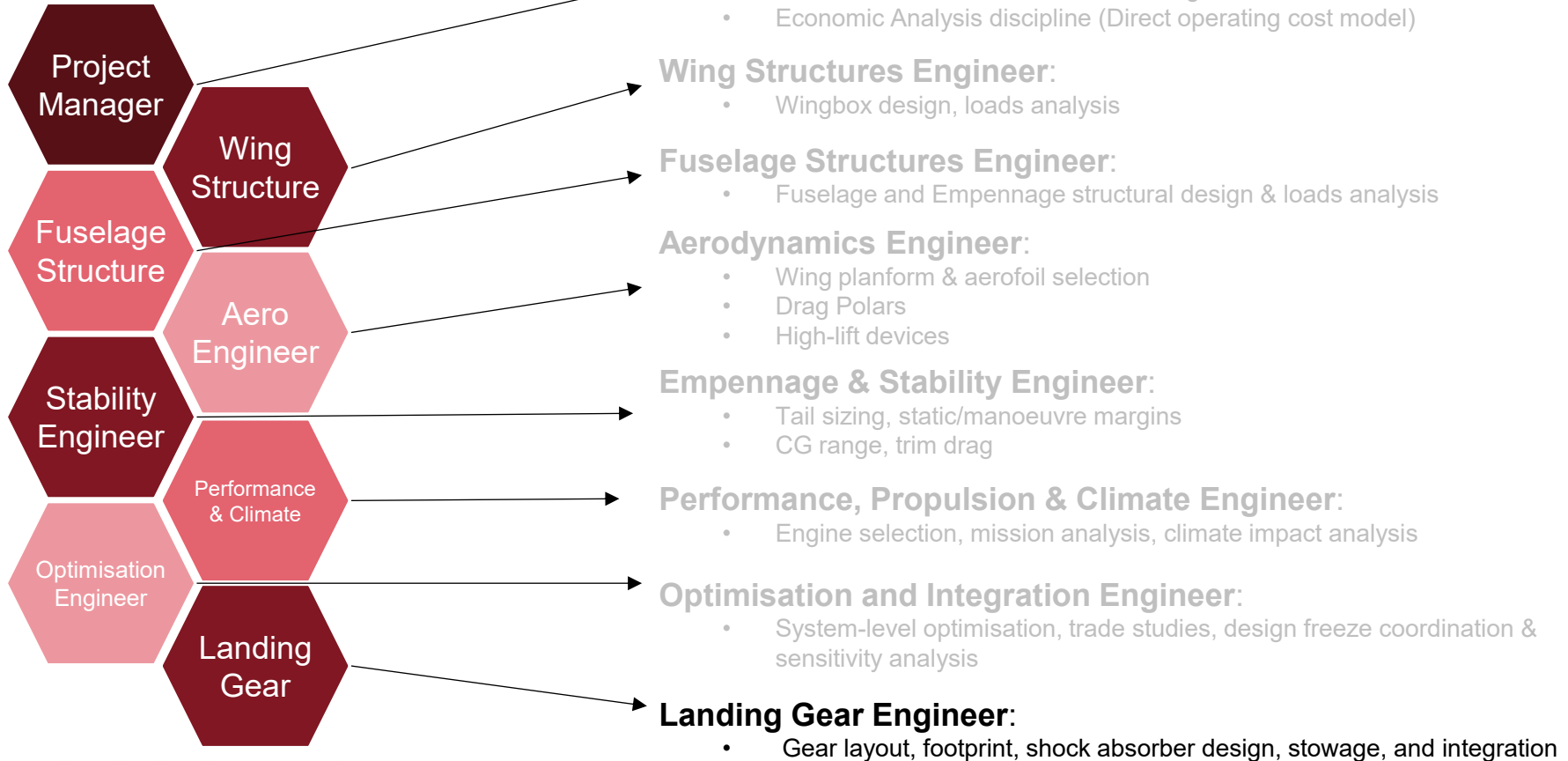
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Team Roles



Team Roles



Questions On Unit Structure, Groups And Roles?

THE DESIGN TASK 2026

A brief overview of the specification document!

Custom Cargo Aircraft for Formula 1

- ~1,200 tonnes of equipment transported to each venue.
- A blend of air, sea, and road freight is used throughout the year.
 - Non-critical components are transported by sea
- Critical components either use air or road freight, depending on the time/distance between adjacent races...

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 MAR 06-08	ROUND 1 MELBOURNE AUSTRALIA	 JUL 24-26	ROUND 13 BUDAPEST HUNGARY
 MAR 13-15	ROUND 2 SHANGHAI CHINA	 AUG 21-23	ROUND 14 ZANDVOORT NETHERLANDS
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 MAY 01-03	ROUND 6 MIAMI MIAMI	 OCT 09-11	ROUND 18 SINGAPORE SINGAPORE
 MAY 22-24	ROUND 7 MONTREAL CANADA	 OCT 23-25	ROUND 19 AUSTIN UNITED STATES
 JUN 05-07	ROUND 8 MONACO MONACO	 OCT NOV 30-01	ROUND 20 MEXICO CITY MEXICO
 JUN 12-14	ROUND 9 BARCELONA - CATALUNYA SPAIN	 NOV 06-08	ROUND 21 SÃO PAULO BRAZIL
 JUN 26-28	ROUND 10 SPIELBERG AUSTRIA	 NOV 19-21	ROUND 22 LAS VEGAS LAS VEGAS
 JUL 03-05	ROUND 11 SILVERSTONE GREAT BRITAIN	 NOV 27-29	ROUND 23 LUSAIL QATAR
 JUL 17-19	ROUND 12 SPA-FRANCORCHAMPS BELGIUM	 DEC 04-06	ROUND 24 YAS ISLAND ABU DHABI

Custom Cargo Aircraft for Formula 1

- F1 aims to be net-zero by 2030.
- A significant proportion of carbon emissions is from air transport
- F1 wants to conduct a feasibility study on developing its own fleet of cargo aircraft to reduce climate impact

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IMPACT 2019

FORMULA 1 ANNOUNCES PLAN TO BE NET ZERO CARBON BY 2030

November 12, 2019

Design Task

- Complete the conceptual design of a custom cargo aircraft for *entry into service in 2040*
- A **fleet** of these aircraft must:
 - Transport the specified cargo across each flyaway leg of the F1 season
 - Ensure freight is delivered within the time constraints of the F1 calendar.
- The chosen aircraft will look to minimise both:
 - Direct operating costs
 - Climate Impact
- The chosen aircraft will be compliant with CS-25

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or?



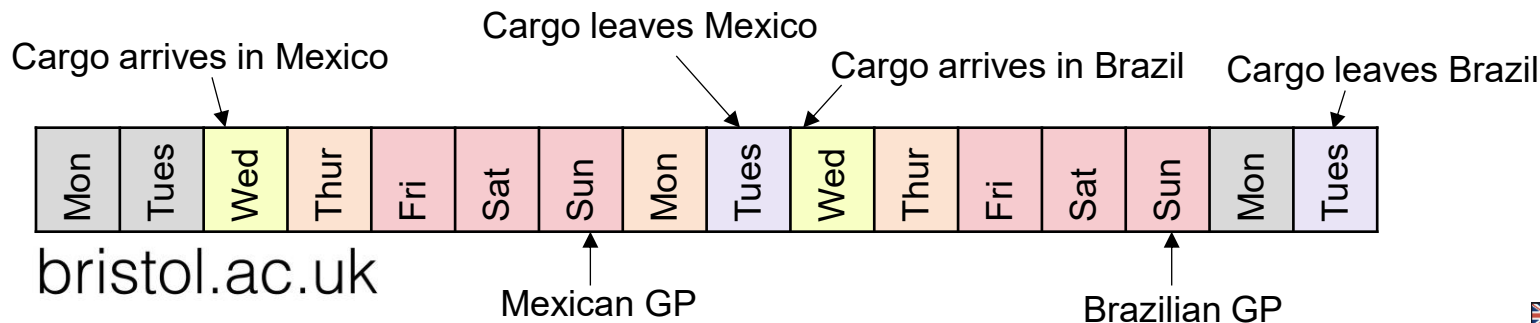
Entry into Service - 2040

- Your aircraft will enter into service (EIS) from 2040. What does that mean for you?
- Your design decisions need to consider technologies and materials that are appropriate for this timeframe
 - Technologies for the 2040 EIS will need to be available earlier to allow a design freeze
- Do customers value “fancy” technologies?
 - Technology choices need to be understood in terms of adding value (\$ € £ etc.)



Flight Schedule

- Use the 2026 calendar as a proxy for 2040
- There are a total of 16 'legs' which utilise air freight
 - Freight starts and ends the season in the UK*
 - Some flights are "empty" if road transport can be used instead
- Some race weekends are back-to-back:
 - The last cargo flight leaves 40 hours after the race start
 - Cargo must arrive at the circuit by Wednesday, 6 am local time
 - There is a time difference between countries



UK	MAR	04-08	ROUND 1: MELBOURNE	AUSTRALIA
CHN	MAR	13-15	ROUND 2: SHANGHAI	CHINA
JPN	MAR	27-29	ROUND 3: SUZUKA	JAPAN
BHR	APR	18-19	ROUND 4: BAHRAIN	BAHRAIN
KSA	APR	17-19	ROUND 5: JEDDAH	SAUDI ARABIA
USA	MAY	01-03	ROUND 6: MIAMI	MIAMI
CAN	MAY	22-24	ROUND 7: MONTREAL	CANADA
MCO	JUN	05-07	ROUND 8: MONACO	MONACO
ESP	JUN	12-14	ROUND 9: BARCELONA - CATALUNYA	SPAIN
AUT	JUN	26-28	ROUND 10: SPIELBERG	AUSTRIA
GBR	JUL	03-05	ROUND 11: SILVERSTONE	GREAT BRITAIN
BEL	JUL	17-19	ROUND 12: SPA - FRANCISCUSWEG	BELGIUM
HUN	JUL	24-26	ROUND 13: BUDAPEST	HUNGARY
NLD	AUG	21-23	ROUND 14: ZANDVOORT	NETHERLANDS
ITA	SEP	04-06	ROUND 15: MONZA	ITALY
ESP	SEP	11-13	ROUND 16: MADRID	SPAIN
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SIN	OCT	09-11	ROUND 18: SINGAPORE	SINGAPORE
USA	OCT	23-25	ROUND 19: AUSTIN	UNITED STATES
MEX	OCT	30-01	ROUND 20: MEXICO CITY	MEXICO
BRA	NOV	04-08	ROUND 21: SAO PAULO	BRAZIL
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UAE	DEC	04-06	ROUND 24: YAS VICINA	ABU DHABI

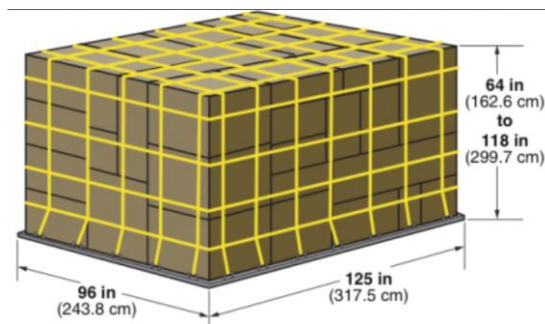
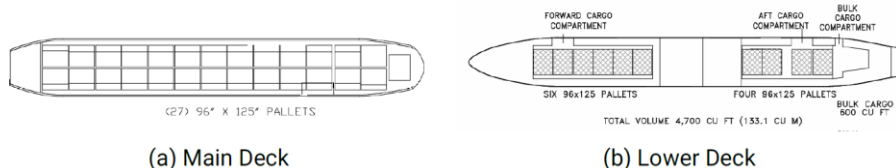
Flight Schedule

Race Name	City	Date (2026)	Race Start Time (Local)	Closest Internatinal Airport	Air Frieight In?	Air Frieight Out?	Notes
Australia	Melbourne	March 6-8	15:00	Melbourne Airport (MEL)	yes	yes	pre-race fleet flies in from UK
China	Shanghai	March 13-15	15:00	Shanghai Pudong International Airport (PVG)	yes	yes	
Japan	Suzuka	March 27-29	14:00	Chubu Centrair International Airport (NGO)	yes	yes	
Bahrain	Sakhir	April 10-12	18:00	Bahrain International Airport (BAH)	yes	yes	
Saudi Arabia	Jeddah	April 17-19	20:00	King Abdulaziz International Airport (JED)	yes	yes	
USA	Miami	May 1-3	16:00	Miami International Airport (MIA)	yes	no	Fleet flies empty to Montreal
Canada	Montreal	May 22-24	14:00	Montréal-Trudeau International Airport (YUL)	no	yes	
Monaco	Monaco	June 5-7	15:00	Nice CÂ'te d'Azur Airport (NCE)	yes	no	Fleet flies empty to Madrid
Spain	Barcelona-Catalunya	June 12-14	15:00	Barcelona El Prat Airport (BCN)	no	no	
Austria	Spielberg	June 26-28	15:00	Vienna International Airport (VIE)	no	no	
Great Britain	Silverstone	July 3-5	15:00	London Heathrow Airport (LHR)	no	no	
Belgium	Spa-Francorchamps	July 17-19	15:00	Brussels Airport (BRU)	no	no	
Hungary	Budapest	July 24-26	15:00	Budapest Ferenc Liszt International Airport (BUD)	no	no	
Netherlands	Zandvoort	August 21-23	15:00	Amsterdam Schiphol Airport (AMS)	no	no	
Italy	Monza	September 4-6	15:00	Milan Malpensa Airport (MXP)	no	no	
Spain	Madrid*	September 11-13	15:00	Adolfo Suárez Madrid-Barajas Airport (MAD)	no	yes	
Azerbaijan	Baku	September 24-26	15:00	Heydar Aliyev International Airport (GYD)	yes	yes	
Singapore	Singapore	October 9-11	20:00	Singapore Changi Airport (SIN)	yes	yes	
USA	Austin	October 23-25	14:00	Austin-Bergstrom International Airport (AUS)	yes	no	Fleet flies empty to Mexico
Mexico	Mexico City	October 30 - November 1	14:00	Mexico City International Airport (MEX)	no	yes	
Brazil	Sao Paulo	November 6-8	14:00	São Paulo Guarulhos International Airport (GRU)	yes	yes	
USA	Las Vegas	November 19-21	22:00	Harry Reid International Airport (LAS)	yes	yes	
Qatar	Lusail	November 27-29	20:00	Hamad International Airport (DOH)	yes	yes	
Abu Dhabi	Yas Marina	December 4-6	17:00	Abu Dhabi International Airport (AUH)	yes	yes	post-race fleet flies to the UK

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Cargo Specification

- For each 'full' leg:
 - ~650 tonnes transported by air
 - Each team transports (assume 12 teams):
 - 13 P6P pallets @ 3 tonnes
 - F1 transports:
 - 4 P6P+ @ 2.5 tonnes (safety cars...)
 - 86 P6P @ 3 tonnes (FIA gear, timing systems)



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JP	ROUND 3: SUZUKA	JAPAN
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Figure 5: Example Deck Layouts for the B777F with 96x125 inch pallets

Getting Started

Top-Level Aircraft Requirements

- Before sketching an aircraft, you **must** understand the specification!
- Design starts with **requirements**, not geometry
- The first step is converting customer requirements into top-level aircraft requirements (**TLARs**)

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Delta Wing



Mid Wing



High Wing



Variable Sweep



Forward Sweep



Inverse Delta Wing

“Carry 246 pallets across the season.”
“Arrive by 6am on Wednesday for each race.”
“Use airports of 2026 calender”

TO

“Max. Payload mass”
“Max. range”
“field length requirements”

Etc...

Customer Versus Aircraft Requirements

- Customer requirements are what F1 cares about
 - Payload (mass, volume, geometry)
 - Mission schedule (timings)
 - Airports
 - Cost (DOC minimisation)
 - Climate impact (CO₂ + non-CO₂)
 - Entry into service (2040 technologies allowed)
- TLARs are what you need to design an aircraft
 - **Payload Requirements**
 - Mass/geometry, loading constraints
 - **Performance Requirements**
 - Range, cruise Mach number, take-off/landing distance, climb gradients, ETOPS
 - **Geometric Requirements**
 - Wingspan limits, fuselage size constraints
 - **Operational Requirements**
 - Fuel type, crew mission profile

TLARs are not a Design Decision

- *TLARs describe what the aircraft must do, not how it does it.*
- TLARs do not specify:
 - Wing planform
 - Number of engines
 - Materials
 - High-Lift systems
 - Tail configuration
- Those are design choices made after TLARs are set!

Suggested Starting Tasks

As a team:

- Distil the specification into a list (Excel sheet) of customer requirements
 - This can be used during the project to check you're meeting objectives
 - Range and field lengths (runway lengths) can be extracted from "*Flight Schedule*"
- Convert this into a set of TLARs
 - Note these could be considered at fleet level or aircraft level



TLAR	Value	Unit
Crew	4	-
Max. Payload	N	tn
Max. Range	M	Km
Cruise Mach Number	x.xx	-
Take-off balance field length	x.x	km
Landing Distance	x.x	km
...	...	...

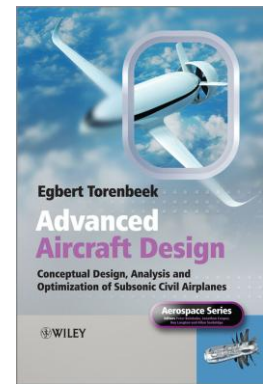
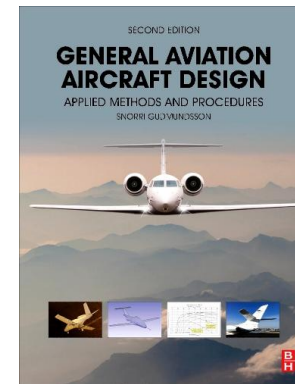
Not exhaustive!

Final Notes

will

- In the short term, you may feel overwhelmed by information
 - That's OK, there is a lot of uncertainty at the start of a design process
- **How can we do well?**
 - The team with the best DOC or lowest climate impact will not necessarily get the best mark
 - We are looking to see how you make decisions and come to justifiable conclusions
 - Explain and justify your decisions, don't just state the end result
- **The specification is fluid:**
 - We have not fully pre-assessed the spec. So we don't know what the "right" answer is, or what is achievable
 - The customer (i.e., me) may not appreciate what they are asking for.
 - If you feel the specification is too restrictive (or not restrictive enough), talk to the customer (me...)
- Lots of good resources for conceptual aircraft design

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<https://bris.on.worldcat.org/oclc/1428781796>

<https://bris.on.worldcat.org/oclc/1282596623>

<https://bris.on.worldcat.org/oclc/842881790>

[Easy Access Rules for Large Aeroplanes \(CS-25\)](#)

Questions?